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Initializing the study

x	Eclipse 100
x	Eclipse 300

Thermal option only

The initial reservoir conditions can be defined in one of four ways:

- The first method assumes the reservoir is in hydrostatic equilibrium. See [Basic hydrostatic equilibration](#).

The other three methods are all forms of non-equilibrium initialization:

- The initial conditions can be read from a RESTART file generated by an earlier run. See ["Restarts"](#).
- The initial conditions can be set explicitly. See ["Enumeration"](#).
- The initial conditions can be calculated by a combination of equilibration and enumeration. See ["Mixed hydrostatic equilibration"](#).

In compositional equation of state (EoS) models, determining whether a single phase is liquid or vapor can be problematic. This issue is discussed in ["Single phase states in EoS compositional models"](#).

Further information on initializing:

- ["Using the Eclipse tracer tracking facility"](#) for details on initializing tracers in Eclipse.
- ["Using the polymer flood model"](#) for details on initializing the Polymer Flood Model in Eclipse.
- ["Injecting solvent into the reservoir"](#) for details on initializing the Solvent Model in Eclipse.

Equilibration

Basic hydrostatic equilibration

For a reservoir containing a single phase fluid, the vertical pressure gradient can be determined from the fluid density. If the reservoir pressure is given at a datum depth, then the fluid density can be calculated at that depth and the pressure in the reservoir can be determined by marching up (and down) the reservoir in small steps, recalculating the density at each step.

The algorithm is more complicated when more than one phase is present. Contact depths (where two phases meet) must be provided, together with the capillary pressure at these depths. In addition sufficient information must be provided to allow the densities of all phases to be calculated as described in ["Density calculations"](#). Starting at the datum depth, the hydrostatic pressure for the "datum" phase can be calculated by marching up and down the reservoir. Pressures in the other phases can then be determined at the contact depths, and then the hydrostatic pressures can be determined throughout the reservoir by marching up and

down again. Once the phase pressures are known, phase saturations can be determined at each depth so that the hydrostatic pressure variation is balanced by the capillary pressure between the phases. Further details are given in ["Calculating the initial conditions"](#).

When two or more phases are present, the algorithm assumes that water is heavier than oil which is heavier than gas. If this condition does not hold, then the algorithm will not work properly and the results will not be as expected. Failure of the condition can be a sign that the fluid has not been properly characterized, or, if heavy oil is being modeled, that the oil may in fact be heavier than water. For more information, see ["Heavy oil"](#).

The hydrostatic equilibration provides tables of pressure and saturation as a function of depth. The equilibration can be performed on a finer grid than that used for the simulation, in which case the resulting saturations are integrated over the cell volumes at the end of the equilibration to provide a starting point for the simulation. In general, the use of a fine grid equilibration will yield a more accurate fluid in place, but may lead to some fluid movement upon starting the simulation; see ["The quiescence option"](#).

The aim of equilibration is to set up a static initial configuration in which the phases present are in equilibrium and in which the inter-block flows are zero. Successful equilibration can be verified by running the simulation without wells to check that no flows occur.

Equilibration regions

The reservoir can be divided into a number of 'equilibration regions' in which hydrostatic equilibrium exists independently of other regions. The number of equilibration regions (NTEQUL) is specified by using the [EQLDIMS](#) keyword in the RUNSPEC section. Cells are associated with these regions using the [EQLNUM](#) keyword. Cells in a given equilibration region should have the same pressure table for their PVT properties, but they can use different saturation tables.

Cells in different equilibration regions should be non-communicating, otherwise they may not be in equilibrium. See ["The threshold pressure option"](#).

Contact and datum depths

For each equilibration region, the fluid contact depths, the capillary pressures at these depths and the pressure at a given datum depth are specified using the [EQUIL](#) keyword.

The datum depth for the pressure can usually lie anywhere in the equilibration region. However, in black oil runs, if dissolved gas or vaporized oil data are required but defaulted, then the datum depth must correspond to the gas-oil contact.

If there is no gas cap, the gas-oil contact should be placed above the top of the reservoir; and likewise if there is no mobile water, the water-oil contact should be placed far enough below the bottom of the reservoir to exclude the transition zone.

In gas-water problems, the water-oil contact and the gas-oil contact should both be placed at the gas-water contact. The gas-water capillary pressure should then be supplied as a function of the water saturation.

The water-oil and gas-oil contacts should also be placed at the same depth for three-phase condensate cases in which the initial state is above the dew point.

It is also possible to have a gas-water contact in a three-phase problem; namely in a gas condensate problem where all the oil is initially vaporized in the gas phase. In this case, the water-oil contact and the gas-oil contact should both be placed at the gas-water contact.

In problems containing a gas-water contact, the gas-water capillary pressure should be supplied in the water saturation tables (keyword [SWFN](#)). For consistency, the capillary pressure data in the gas saturation tables should be set to zero.

Density calculations

Data describing the fluid as a function of depth need to be provided to allow the equilibration algorithm to calculate the fluid density and hence the hydrostatic pressure gradient. The data required depends on the fluid model and is described separately in the sections "[Black oil composition with respect to depth](#)" and "[Initial composition with respect to depth](#)". Unless defaulted, tables must be entered for each [EQLNUM](#) region. The maximum number of depth values in any such table must be set using the keyword [EQLDIMS](#) in the [RUNSPEC](#) section.

In gas-water problems, there is no need to enter any data concerning the nonexistent oil phase.

Black oil composition with respect to depth

Dissolved gas

In black oil models containing dissolved gas, the dissolved gas concentration R_s or the bubble point pressure P_b may be entered as a function of depth using the keywords [RSVD](#) and [PBVD](#) respectively. A table must be entered for each [EQLNUM](#) region unless the data is defaulted altogether. At any position in the reservoir, the R_s value must be less than or equal to the saturated R_s value at the local pressure.

In runs with dissolved gas, if both the [RSVD](#) and [PBVD](#) keywords are omitted, the dissolved gas concentration in undersaturated oil is set everywhere to be equal to the saturated R_s value at the gas-oil contact. In this case, the datum depth for the equilibration must be set at the at the gas-oil contact, so that the saturated R_s value can be calculated there.

Vaporized oil

In gas condensate problems, the vaporized oil concentration R_v or the dew point pressure P_d may be entered as a function of depth using the keywords [RVVD](#) and [PDVD](#) respectively. A table must be entered for each [EQLNUM](#) region unless the data is defaulted altogether. At any position in the reservoir, the R_v value must be less than or equal to the saturated R_v value at the local pressure.

In runs with vaporized oil, if both the [RVVD](#) and [PDVD](#) keywords are omitted, the vaporized oil concentration in undersaturated gas is set everywhere to be equal to the saturated R_v value at the gas-oil contact. In this case, the datum depth for the equilibration must be set at the gas-oil contact, so that the saturated R_v value can be calculated there.

Initial composition with respect to depth

In Eclipse 300 compositional mode, an extra complication exists. As well as ensuring that the initial saturations are in equilibrium, the thermodynamic equilibrium between phases must be considered. Three methods exist for setting up the initial compositions, which usually vary with depth:

1. For single hydrocarbon phase initial states, such as condensates above the dew point, or oils above the bubble point, or for "supercritical" reservoirs in which a smooth transition exists between the oil and gas properties and the saturation pressure is less than the fluid pressure throughout the depth range:

Use [ZMFVD](#) to specify the initial composition with respect to depth. The gas-oil contact may be set above or below the reservoir if the single phase hydrocarbon is an oil or gas. In the supercritical case the gas-oil contact depth may be set to a point within the reservoir. The simulator then sets the phase

identification method to produce a phase change at this depth. For a condensate field the gas-oil contact depth is set to the oil-water contact depth. This identifies the study as a condensate to the simulator.

2. For cases in which there is a gas-oil contact within the reservoir, and the vapor composition is known:

Use ZMFVD to specify the initial vapor composition. A retrograde dew point pressure calculation is performed for the vapor and the pressure is set to this at the contact. The composition of the liquid in equilibrium with the saturated vapor is used below the contact. As pressure increases with depth this is undersaturated.

Alternatively, COMPVD may be used to specify a compositional variation in the liquid as well as in the vapor. However, the composition specified at the gas-oil contact must be that of a gas.

3. For cases in which there is a gas-oil contact within the reservoir, and the liquid composition is known:

Use ZMFVD to specify the initial oil composition. A bubble point pressure calculation is performed for the liquid and the pressure is set to this at the contact. The composition of the vapor in equilibrium with the saturated liquid is used above the contact. As pressure decreases above the contact this may yield an oil phase upon flashing. If so, only the vapor is used, so that saturated vapor only exists above the contact.

Alternatively, COMPVD may be used to specify a compositional variation in the vapor as well as in the liquid. However, the composition specified at the gas-oil contact must be that of an oil.

After the calculations described in cases 2 and 3 above, the oil composition is known below the gas-oil contact, and the vapor composition is known above the gas-oil contact. Oil may be present above the contact, and vapor below, depending on the capillary pressure. Therefore it is necessary to determine the composition of these phases:

- If there is any vapor below the contact then it will be in equilibrium with the oil below the contact. Therefore a bubble point calculation is performed on the oil below the contact. If the reservoir pressure is below the oil bubble point then the vapor composition can be set equal to the composition of the vapor in equilibrium with the oil at the reservoir pressure. If however the reservoir pressure is above the oil bubble point then the vapor composition will be set equal to the bubble point vapor composition. In this case the oil and vapor will not be in equilibrium, and as a result - if vapor is in fact present at these depths - the algorithm will fail to match the required compositions and saturations.
- If there is any oil above the contact then it will be in equilibrium with the vapor above the contact. Therefore a dew point calculation is performed on the vapor above the contact. If the reservoir pressure is above the vapor dew point then the oil composition can be set equal to the composition of the oil in equilibrium with the vapor at the reservoir pressure. If however the reservoir pressure is below the vapor dew point then the oil composition will be set equal to the dew point oil composition. In this case the oil and vapor will not be in equilibrium, and as a result - if oil is in fact present at these depths - the algorithm will fail to match the required compositions and saturations.

The calculated saturation pressures, and oil and vapor compositions and phase saturations can be output as a function of depth by specifying the SOLVD mnemonic with the RPTSOL keyword. The compositions and saturations will match the actual values in the reservoir if the oil and vapor are in equilibrium. However they will fail to match when the oil and vapor are not in equilibrium which can occur if the reservoir pressure is above the oil bubble point or below the vapor dew point. In these cases the specified composition (COMPVD or ZMFVD) should be adjusted so that the fluid is two phase at the reservoir pressures.

The three cases above correspond to the initialization type specified by the tenth argument of the EQUIL keyword. In addition, the eleventh argument may be set to 1 to suppress the setting of the pressure at the

contact to the saturation pressure at the contact in options 2 and 3 above. This yields a system that is not in initial equilibrium, but honors the specified pressure.

Calculating the initial conditions

Within each equilibration region, the calculation is performed in two stages. The first stage sets up an internal table of phase pressures (P_w, P_o, P_g), and composition (or R_s and R_v for black oil) against depth. The second stage interpolates this table and uses the saturation functions to obtain the fluid conditions in each grid block in the region.

In the first stage, depth points for the internal tables are spaced equally throughout the region. The number of depth points is set using the keyword `EQLDIMS` in the `RUNSPEC` section. The value should be increased when finer definition is required, for example in problems where the oil zone occupies only a small fraction of the overall reservoir thickness.

Between each pair of depth points in the table, the pressure gradient of each phase is calculated iteratively using a density consistent with the average pressure within the depth step. If the datum depth lies in the oil zone, the oil pressure values are calculated first, stepping from the datum depth up to the top and down to the bottom of the equilibration region. The water pressure on the oil-water contact can now be obtained, and the water pressure values are calculated. Similarly the gas pressure values are calculated starting from the gas-oil contact.

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Depth table data can be reported using the `SOLVD` mnemonic in the `RPTSOL` keyword.

In the second stage of the equilibration calculation, the local fluid conditions are determined in each grid block in the equilibration region. The internal table is interpolated to obtain the values of phase pressures (P_w, P_o, P_g) and composition (or R_s and R_v) in each grid block. Different interpolation methods can be used to improve the accuracy, see "[Accurate fluids in place calculation](#)".

The water saturation is determined by inverse look-up of the water capillary pressure table (entered with keyword `SWFN` or `SWOF`) for the grid block, such that

$$P_{cow}(S_w) = P_o - P_w \quad \text{Eq. 5.1}$$

If $P_o - P_w$ exceeds the highest capillary pressure value in the `SWFN/SWOF` table (corresponding to the lowest saturation value S_{wmin}), the water saturation is set equal to S_{wmin} .

If $P_o - P_w$ is less than the lowest capillary pressure in the table (corresponding to the highest saturation value S_{wmax}), the water saturation is set equal to S_{wmax} and the oil pressure is adjusted to follow the water pressure gradient.

The gas saturation is similarly determined by inverse look-up of the gas capillary pressure table (keyword `SGFN`, `SGOF` or `SLGOF`) for the grid block, such that

$$P_{cog}(S_g) = P_g - P_o \quad \text{Eq. 5.2}$$

If $P_g - P_o$ is less than the lowest capillary pressure value in the table (corresponding to the lowest gas saturation value S_{gmin}), the gas saturation is set equal to S_{gmin} .

If $P_g - P_o$ exceeds the highest capillary pressure in the table (corresponding to the highest gas saturation value S_{gmax}), the gas saturation is set equal to S_{gmax} and the oil pressure is adjusted to follow the gas pressure gradient.

Consistency and overlapping transition zones

Since the water and gas saturations outside the transition zones are set to the respective end-point values in the saturation function tables, the end-points of the two tables must be consistent. Normally the lowest gas saturation in the [SGFN](#) or [SGOF](#) table should be zero (so that $S_g = 0$ below the gas transition zone), and the highest gas saturation in this table should be $S_{gmax} = 1 - S_{wmin}$ (so that $S_o = 0$ above the gas transition zone). The lowest water saturation value in the [SWFN](#) or [SWOF](#) table should be the connate value (so that $S_w = S_{wco}$ above the water transition zone), and the highest water saturation in this table should be 1.0 (so that $S_w = 1$ below the water transition zone). The consistency requirements between saturation tables are detailed in "[Saturation functions](#)". If the consistency requirements are not met, $S_{gmax} < 1 - S_{wmin}$, then not only will $S_o > 0$ above the gas transition zone, it may even increase. If the transition zones overlap, then the water saturation will decrease above the gas-oil contact. The gas saturation will increase until it reaches its maximum value. Above this point, as the water saturation continues to decrease, the oil saturation will increase.

In cases where the water-oil and gas-oil transition zones closely overlap, equilibration may produce negative oil saturations. For example, where the water-oil transition zone extends above the gas-oil transition zone, the water saturation ($S_w > S_{wmin}$) and the gas saturation ($S_g = S_{gmax}$) add to more than 1. If this occurs, the water and gas saturations are recalculated from the gas-water capillary pressure, which is taken as the sum of the water-oil and gas-oil capillary pressures

$$P_{cow}(S_w) + P_{cog}(S_g = 1 - S_w) = P_g - P_w \quad \text{Eq. 5.3}$$

Heavy oil

Eclipse 300 Thermal

If oil is heavier than water (based on the composition provided at the datum depth) and if the water-oil contact lies above the reservoir, then the reservoir will be initialized with heavy oil with the datum pressure being interpreted as the pressure of the oil phase. A non-zero value for [item 217](#) in `OPTIONS3` can be used to restore the pre-2010.1 behavior for thermal heavy oil equilibration.

Accurate fluids in place calculation

There is a choice of methods to calculate the initial fluid in place values (the initial oil, water and gas saturations) in each grid block. These are selected using the ninth argument of the [EQUIL](#) keyword, which specifies the number of sub-layers (N) used to obtain the initial saturations.

Center-point equilibration (N=0)

The simulator sets the fluid saturations in each grid block according to the conditions at its center. This is the fastest method, and yields a stable initial state since the phase pressure differences in the simulation are also taken between cell centers. But it is the least accurate method, particularly in cases where the fluid contact passes through large grid blocks.

Horizontal fine grid equilibration (N<0)

The top and bottom halves of each grid block are each divided into -N layers of equal thickness, and the saturations are determined locally in each layer. The phase saturations for the block are set equal to the average of the saturations in each layer. This option provides a more accurate calculation of the fluids in place, but may yield a solution that is not completely stable in the initial state.

Tilted block fine grid equilibration ($N>0$)

This option is similar to the previous one, but it takes into account the slope of each grid block. The top and bottom faces of the blocks are treated as planes that are tilted about their central points. If the `NEWTRAN` keyword has been entered, the tilt is determined from the block corner point depths. Otherwise the faces are tilted towards the centers of the corresponding faces of the neighboring blocks within the equilibration region. The top and bottom halves of the grid blocks are each divided into N horizontal layers of equal thickness, but the thickness of the layers in the top half is generally different from the thickness of the layers in the bottom half. This is because generally the distance from the block center line to the highest corner is not the same as the distance from the block center line to the lowest corner, if the upper and lower faces have different tilts. The phase saturations in each block are calculated as a weighted average of the saturations in the $2N$ layers, weighted according to the area of each layer that is enclosed within the block multiplied by the layer thickness. This option provides the most accurate calculation of fluids in place, but may yield a solution which is not completely stable in the initial state.

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With fine grid equilibration there is a redistribution of fluids between grid blocks near the contacts when the simulation begins, which occurs independently of any external driving force (wells for example). The reason is that a steady state solution on the fine equilibration grid (in which each block is subdivided into several layers) is not necessarily a steady state solution on the coarser simulation grid. If the redistribution of fluids produces a significant transient when the simulation is started, you can overcome this by setting the 'quiescence switch' (the switch `QUIESC` in keyword `EQLOPTS` in the `RUNSPEC` section). If this switch is on, modifications are applied to the phase pressures to make the initial solution a true steady state. These pressure modifications are applied for the duration of the run.

Eclipse 100

If the vertical equilibrium option is used (keyword `VE` in the `RUNSPEC` section), an exact VE equilibration method can be chosen by selecting center-point equilibration. Instead of the center-point calculation, the option calculates the fluid saturations in each grid block directly from the contact depths. It employs the same relationship between saturations and contact depths that is subsequently used for the simulation. This is the recommended option in VE runs.

The threshold pressure option

If the reservoir contains two or more communicating equilibration regions having different pressures or contact depths, fluid flows between these regions immediately the simulation begins. The 'quiescence switch' only produces local steady state conditions within each separate equilibration region. Global quiescence, however, can be achieved by using the threshold pressure facility. The facility prevents flow between adjacent equilibration regions from occurring until the potential difference exceeds a specified threshold value. Thereafter the potential difference for flow across equilibration region boundaries is reduced by the appropriate threshold value.

The Threshold Pressure facility is enabled by including the switch `THPRES` in keyword `EQLOPTS` in the `RUNSPEC` section. Each interface between adjacent equilibration regions then has to have its own threshold pressure. If the threshold pressure is also irreversible (the switch `IRREVERS` in keyword `EQLOPTS` in the `RUNSPEC` section), each interface has two threshold pressures, one for flow in each direction. The threshold pressure values are set using the keyword `THPRES`. Values can be defaulted to the maximum potential difference between communicating cells across the interface (that is, just sufficient to prevent flow occurring across the interface before the reservoir is perturbed).

The quiescence option

Eclipse 100

When using the fine grid equilibration algorithm to achieve a more accurate fluids-in-place calculation, the initial conditions may no longer be in hydrostatic equilibrium for each phase. This is because the

saturations are now taken as average values over a grid block rather than the values at cell centers. The values of pressure in the fine grid equilibration are still taken from block center values.

The Quiescence option, set using the [EQLOPTS](#) keyword, achieves hydrostatic equilibrium for flows of each phase. It modifies the initial (oil phase) pressure P and introduces cell-dependent modifiers $PMODW$ and $PMODG$ to the water and gas phase pressures such that

$$P_{wat} = P - P_{cow} + PMODW \quad \text{Eq. 5.4}$$

and

$$P_{gas} = P + P_{cog} + PMODG \quad \text{Eq. 5.5}$$

The phase pressure modifications $PMODW$ and $PMODG$ are determined to achieve quiescence at initial conditions and are then applied throughout the simulation. These can be thought of as frozen pseudo-capillary pressure modifications. The quiescent pressures are constructed from the initial tables of phase pressure versus depth, p_{oil} , p_{gas} and p_{wat} over the depth of the field. Then, in the absence of capillary pressures, $p_{oil} = p_{gas}$ at the oil-water contact and $p_{oil} = p_{gas}$ at the gas-water contact.

The (oil phase) pressure in each grid block center, P is modified by

$$P = \max(P, p_{wat} - (\tilde{\rho}_{wat} - \tilde{\rho}_{oil})H/2, p_{gas} - (\tilde{\rho}_o - \tilde{\rho}_g)H/2) \quad \text{Eq. 5.6}$$

where

H is the height of a cell, and

ρ denotes the phase gravity density.

The water and gas phase pressure modifications are then determined from

$$PMODW = \text{sign}(p_{wat} - P + P_{cow}) \times \min(|p_{wat} - P + P_{cow}|, (\tilde{\rho}_w - \tilde{\rho}_o)H/2) \quad \text{Eq. 5.7}$$

and

$$PMODG = \text{sign}(P + P_{cog} - p_{gas}) \times \min(|P + P_{cog} - p_{gas}|, (\tilde{\rho}_o - \tilde{\rho}_g)H/2) \quad \text{Eq. 5.8}$$

These phase pressure modifications are chosen such that the water phase pressure P_{wat} approximately follows the hydrostatic water pressure curve p_{wat} in the presence of mobile water, and likewise so that the gas phase pressure P_{gas} approximately follows the hydrostatic gas pressure curve p_{gas} in the presence of mobile gas.

Note that even for a model with zero capillary pressures, the phase pressure modifications is not necessarily zero. The phase pressure modifications can be reported using the mnemonic 'QUIESC' in the [RPTSOL](#) keyword.

Reservoir temperature

The reservoir temperature must be specified in two different types of simulations:

- Compositional cases, where the temperature is constant in each cell (though it can vary with depth), but is needed to determine the fluid properties.
- Thermal or Temperature option cases, where the temperature can change with time. The initial temperature must be specified to determine the initial energy in the reservoir.

Eclipse 300	In compositional cases, the constant reservoir temperature can be specified by the RTEMP or TEMPVD keywords in the PROPS section.
Eclipse 300 Thermal	In Thermal cases, the initial reservoir temperature can be set using the RTEMPVD keyword in the SOLUTION section. The PROPS section keywords RTEMP or TEMPVD can be used as an alternative. Where temperature keywords are specified in both the PROPS and SOLUTION sections, the SOLUTION section keywords will be used.
Temperature option	In Temperature option cases, the initial reservoir temperature can be set using the RTEMPVD keyword in the SOLUTION section, or the RTEMP and RTEMPA keyword in both the PROPS and SOLUTION sections. In Eclipse 100, the keywords RTEMP and RTEMPA are accepted in the PROPS section only if the EQUIL keyword is defined in the SOLUTION section. Non-equilibrium initializations require these keywords to appear in the SOLUTION section only.

Phase pressures in Eclipse 300

By default, oil and gas phase pressures are derived from the pressure solution variable P using the formula

$$P_{oil} = P - P_{cog} \times \left(\frac{S_g}{S_o + S_g} \right) \quad \text{Eq. 5.9}$$

and

$$P_{gas} = P + P_{cog} \times \left(\frac{S_o}{S_o + S_g} \right) \quad \text{Eq. 5.10}$$

where

P_{cog} is the gas-oil capillary pressure,

S_o and S_g are the oil and gas saturations respectively.

This P_c splitting technique is performed to enable better convergence and property smoothness above the critical point. It may be disabled using keyword [FORMOPTS](#), option [NOPCSPT](#). When reporting initialization data using the [RPTRST](#) or [RPTSOL](#) keywords, the pressures are defined as

PRESSURE is the solution variable P which is derived from the phase pressures calculated during the [EQUIL](#) hydrostatic equilibration and reported with the [SOLVD](#) option in [RPTSOL](#).

POIL is the oil phase pressure P_{oil} defined in equation 5.9

PGAS is the gas phase pressure P_{gas} defined in equation 5.10

Note: The P_c splitting technique applies only to the Eclipse 300 compositional mode.

Equilibration keywords

RUNSPEC

- [EQLDIMS](#) Dimensions of equilibration tables

- [EQLOPTS](#) Options for equilibrations

REGIONS

- [EQLNUM](#) Equilibration region numbers

SOLUTION

Restriction	Keyword	Description
	COMPVD	Composition versus depth table
	EQUIL	Datum depth and pressure, contact depth and capillary pressures, equilibration parameters
	PBVD	Bubble point P_b versus depth table for black oil
	PDVD	Dew point p_d versus depth table for black oil
	RSVD	R_s versus depth table for black oil
Temperature Option	RTEMPA	Initial temperature
Temperature or Thermal Options	RTEMPVD	Initial temperature versus depth table
	RVVD	R_v versus depth table for black oil
	TVDP	Tracer versus depth table
Eclipse 300	WMFVD	Water composition versus depth table.
Eclipse 300	ZMFVD	Composition versus depth table.

Enumeration

In some cases it may be required to start a study from a point in existing production, where a restart cannot be used. For example, when starting a compositional simulation using the results of a black oil model. As flow patterns have developed, this cannot be done by equilibration. Enumeration allows the initial solution to be specified directly. Three sets of data are required for all simulations:

- The pressure:

[PRESSURE](#), or, for Eclipse 100, [PRVD](#).

- Saturations:

[SGAS](#) and [SWAT](#)

or [SGAS](#) and [SOIL](#)

or [SOIL](#) and [SWAT](#).

Note:

In Eclipse 300, in models with the [SOR](#) option active, using the optional keyword [ROMF](#) will change the way the saturations are calculated. Please refer to "[SOR Initialization](#)" for more details.

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- Composition:

For a black oil simulation with dissolved gas, either [PBUB](#) or [RS](#).

For a black oil simulation with vaporized oil, either [PDEW](#) or [RV](#).

For a compositional simulation with two hydrocarbon phases, both [XMF](#) and [YMF](#).

For a compositional simulation with only a single hydrocarbon phase, [ZMF](#).

The above three sets of data are sufficient for most simulations. However, special options may require additional data. For example:

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- Hydrocarbon components in water:

[AMF](#)

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- Water components in water:

[WMF](#)

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- Solid model

Solid composition:

[SMF](#)

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- The temperature:

[TEMPI](#)

There are several alternative ways of entering this data:

- Like `GRID` and `EDIT` section keywords, all the enumeration keywords may also be set using [EQUALS](#), [ADD](#) and [MULTIPLY](#) applied to boxes of cells.

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- Data can be read directly from a flexible restart file using the [GETSOL](#) keyword. The format of this is:

```
GETSOL
PRESSURE BASE1 10 /
SWAT BASE1 10 /
/
```

This would imply that the pressure and water saturation were to be obtained from the 10th report step of the `RESTART` file with root name `BASE1`.

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- Instead of [XMF](#) and [YMF](#), tables of composition with respect to pressure can be entered using the [XMFVP](#) and [YMFVP](#) keywords. These tables would usually be generated by the PVT *i* package from constant volume depletion or differential liberation experiments, and would specify a set of equilibrium liquid and vapor compositions as a function of pressure. This should help ensure that the chosen liquid and vapor are close to phase equilibrium and avoid large transient effects that occur when the initial phase are not in equilibrium. This is discussed further in "[Specifying the initial state of a run in Eclipse 300](#)".

Specifying the initial state of a run in Eclipse 300

The initial state of a cell is defined by P , S_w and S_g . In the black oil case the bubble point may also be defined, while in the compositional case the mole fractions of the oil and gas must be specified, (x_i and y_i , $i = 1, \dots, N_{comps}$). The condition that the liquid and vapor be in equilibrium, in the compositional case, is:

$$f_{iL}(P, x) = f_{iV}(P, y) \quad \text{Eq. 5.11}$$

Note that if this is satisfied, the liquid and vapor may be present in any amounts. The mole fractions of oil and gas present may be established by obtaining their molar densities, and then the total mole fractions are obtained by:

$$z_i = x_i L + y_i V \quad \text{Eq. 5.12}$$

with

$$L = \frac{S_o b_o^m}{S_o b_o^m + S_g b_g^m} \quad \text{Eq. 5.13}$$

where b_o^m and b_g^m are the oil and gas molar densities.

As long as equation 5.11 above is satisfied, flashing a system with total composition z_i then reproduces the original values of L , V , x_i , y_i , S_o and S_g .

If only pressure and saturation data is available, this presents a problem. A single oil and a single gas composition may be specified. This, however, can only be in equilibrium at a single pressure, and equation 5.11 is not satisfied at any other pressure. The result is that the initial flash yields different liquid and vapor mole fractions and saturations from those specified, and there may well be serious convergence problems in the initial state. Even if these are resolved, the system generally no longer has the required saturation values.

Apart from the oil and gas compositions, the inter phase equilibrium condition only involves the pressure P . If a set of equilibrium x_i and y_i values are available parameterized by pressure, it would be possible to start cells at any pressure in equilibrium.

Such a parameterized set of compositions can be obtained from a constant volume depletion or differential liberation experiment. In these, for a condensate or an oil respectively, vapor is extracted at each pressure step that is in equilibrium with the liquid. Further, these experiments attempt to simulate the depletion process that occurs in the reservoir, so are likely to yield representative composition versus pressure tables. Such an experiment can be performed in the PVT i program, and the resulting [XMFVP](#) and [YMFVP](#) (x_i mole fraction and y_i mole fraction versus pressure) tables used as input to Eclipse 300. The pressure steps must be small to avoid interpolation errors in the table, and the range of pressures must span that occurring in the reservoir in initial state.

In some areas of the reservoir the oil may be undersaturated. In this case a bubble point exists, the pressure at which the cell would become two-phase. Only oil exists in this cell, and if it is given a composition taken from the equilibrium oil composition at P_b rather than P , it must be in equilibrium with gas at P_b , and so have the correct saturation pressure.

Using this method, it is possible to set up a set of representative compositions such that inter phase equilibrium is observed for any specified cell pressures.

Enumeration keywords

PROPS

The following keywords can only be used in Eclipse 300.

- [XMFVP](#) Liquid composition versus pressure table
- [YMFVP](#) Vapor composition versus pressure table

SOLUTION

Restriction	Keyword	Description
Eclipse 300	AMF	Initial water compositions
Eclipse 300	GETSOL	Obtain initial solution array from a restart file.
	PBUB	Initial bubble point pressures
	PDEW	Initial dew point pressures
	PRESSURE	Initial pressures
Eclipse 100	PRVD	Initial pressures versus depth tables.
	RS	Initial solution gas-oil ratios
	RV	Initial vapor oil-gas ratios
	SGAS	Initial gas saturations
Eclipse 300	SMF	Initial solid compositions. T
Eclipse 300	SOIL	Initial oil saturations
Eclipse 300	SWAT	Initial water saturations
Eclipse 300	TEMPI	Initial temperatures.
Eclipse 300	XMF	Initial oil compositions.
Eclipse 300	YMF	Initial gas compositions
	ZMF	Initial oil compositions

Mixed hydrostatic equilibration

Thermal

An additional initialization method has been added for Thermal simulations. This option should be triggered using items 1 and 2 of the **EQUIL** keyword, together with saturation arrays (**SOIL**, for instance) and the **NEI** keyword.

For each equilibration region, the datum depths and the pressure at a given datum depth are specified in order to determine the initial pressure map for the simulation model. The number of depth points in this table is set using the keyword **EQLDIMS** in the **RUNSPEC** section. The depth points are spaced equally throughout the region. The software algorithm sets up an internal table of pressures against depths. Between each pair of depth points in the table, the pressure gradient is calculated iteratively using a density (assuming an oil phase) consistent with the average pressure within the depth step. The pressure values are calculated by stepping from the datum depth up to the top and down to the bottom of the equilibration region. The pressure and temperatures are then determined in each grid block, in the equilibration region, when the internal table is interpolated to obtain these values at the grid block center depth.

The **NEI** keyword provides a sample hydrocarbon composition for each equilibration region which is needed to determine non-equilibrium initialization composition tables internally by generating consistent oil and gas compositions for each cell. This is achieved by flashing to the cell pressure - effectively a constant composition expansion. The simulator tries to match the oil, water and gas saturations in each equilibration region, and if the specified compositions cannot match the phase saturations, then the components that have the highest gas phase mole fraction have their compositions adjusted.

If the **NEI** sample provided is only for an oil composition, then the pressure calculated for any gas phase present will not be defined efficiently. An alternative method is to use the **XMFVP** / **YMFVP** or **XMF** / **YMF** keywords.

A table of depth values against initial pressure and temperature values will be automatically output to the PRINT file under the header "MH_EQUIL" (Pressure with depth (Mixed Hydrostatic Equilibration)).

This initialization method should not be mixed with the standard equilibration method, otherwise an error message will be issued.

Single phase states in EoS compositional models

Eclipse 300

Distinguishing whether a single phase fluid is a liquid or a vapor is not always straightforward, and is sometimes arbitrary, for example if the fluid is supercritical, $P > P_c$ or $T > T_c$. If the fluid properties (for example density, viscosity) are determined correctly, then it might be thought that determining whether the fluid is liquid or vapor is only a matter of labelling / reporting and has no effect on the simulation. However, the label is important, since it effects wells controlled by liquid or gas rates.

To determine a single phase state at (P, T) correctly, the bubble point $P_{bub}(T)$, dew point $P_{dew}(T)$ or critical point (P_c, T_c) may be needed. These cannot be calculated quickly, and to do so for every flash would slow down the simulation significantly. Therefore an approximate critical temperature is calculated for the mixture using the Li formula:

$$T_c^{mix} = F_{Li} \cdot \frac{\sum_j T_{cj} \cdot V_{cj} \cdot z_j}{\sum_j V_{cj} z_j}$$

where

z_j is the mole fraction of component j in the fluid,

T_{cj} is the critical temperature of component j , given by the [TCRIT](#) keyword,

V_{cj} is the critical volume of component j , given by the [VCRIT](#) keyword,

F_{Li} is a tuning factor.

Single phase states are then labelled as liquid if $T_c^{mix} > T$ and gas if $T_c^{mix} < T$.

The tuning factor F_{Li} can be set using the [FACTLI](#) keyword. Alternatively the simulator will calculate a value of F_{Li} to ensure where possible that any single phase cell in the reservoir has the correct user-specified liquid or vapor state.